

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES	
10							*****VALVES*****			812	NIPPLE TO BE CUT FROM SS SCH XXS PIPE.
20										816	COMPACT FLANGES CAN BE USED FOR TIE-IN PURPOSE ONLY TO EXISTING PIPING AND EQUIPMENT.
20							***** MODULAR VALVES *****				
30	U04		MODULAR DBB	K06AC02	0.5	1.5	MODULAR VALVE DB&B BALL TYPE 600# RF LTCS A350 LF2 CL.1 316 SS TRIM METAL LO - FULL SIZE TYPE 2BV	0	0	B1	ALL BOLTS ARE METRIC DIA.
40	U04		MODULAR DBB	K06AC06	2	2	MODULAR VALVE DB&B BALL TYPE 600# RF LTCS A350 LF2 CL.1 316 SS TRIM METAL LO - FULL SIZE TYPE 2BV	0	0	B6	7/8" X 145MM BOLTS ONLY TO BE USED FOR 1 INCH 1500# PERAR BALL VALVES ITEM CODE B15AS22
50	U04		MODULAR DBB	K06AC06	2	2	MODULAR VALVE DB&B BALL TYPE 600# RF LTCS A350 LF2 CL.1 316 SS TRIM METAL LO - FULL SIZE TYPE 2BV	0	0	FA8	USE THREADED FLANGES FOR INSTRUMENT CONNECTIONS ONLY.
60	U04		MODULAR DBB	M06AC02	0.5	1.5	MODULAR VALVE DB&B BALL TYPE 600# RF LTCS A350 LF2 CL.1 316 SS TRIM METAL LO - CONTROLLED BORE TYPE 2BVR	0	0	GA6	SPECIAL GASKET ONLY FOR USE AS SPECIFIED IN ENG-200-117
70	U04		MODULAR DBB	M06AC06	2	2	MODULAR VALVE DB&B BALL TYPE 600# RF LTCS A350 LF2 CL.1 316 SS TRIM METAL LO - CONTROLLED BORE TYPE 2BVR	0	0	SB6	USE JACKSCREWS FOR PADDLE BLANKS/SPACERS AND FLANGE SPREADERS FOR SPECTACLE BLINDS.
80	U04	VAW	MODULAR DBB	M06EC02	1	1	MODULAR VALVE DB&B BALL TYPE 600# RF LTCS A350 LF2 CL.1 316 SS TRIM METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR	3	3	UJ2	USE 1 1/2IN WITH THERMOWELL CONNECTIONS ONLY & 3/4IN (IF SHOWN) FOR DBB PRESSURE CONNECTIONS WITH CHEMICAL SEALS.
90	U04	VAW	MODULAR DBB	M06EC02	0.75	1	MODULAR VALVE DB&B BALL TYPE 600# RF LTCS A350 LF2 CL.1 316 SS TRIM METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR	1.5	2		
100	U04	VAW	MODULAR DBB	M06EC02L	0.75	0.75	MODULAR VALVE DB&B BALL TYPE 600# RF LTCS A350 LF2 CL.1 316 SS TRIM METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR	1.5	1.5	UK9	DO NOT USE FOR ORIFICE FLANGE TAP ASSEMBLIES.
110							***** RISING STEM BALL VALVES *****			VA5	USE THESE VALVES ONLY WHERE SPECIFIED ON PIPING AND INSTRUMENT DIAGRAMS
110										VAW	VALVE ONLY TO BE USED IN ACCORDANCE WITH DOCUMENT L-ST-2055 - VENT DRAIN AND INSTRUMENT CONNECTION DETAILS.
120		VA5	BALL	R06AC06	0.75	6	BALL RISING STEM 600# RF LTCS A352 LCC 316SS TRIM METAL RB HO	0	0	VC1	SPRING PISTON CHECK VALVES ARE PREFERED IN THE HORIZONTAL POSITION, BUT MAY BE USED IN THE VERTICAL POSITION WITH ENGINEERING APPROVAL.
130	U04	VA5	BALL	R06AC08	8	10	BALL RISING STEM 600# RF LTCS A352 LCC 316SS TRIM METAL RB GO	0	0		
140							***** BUTTERFLY VALVES *****				
140											
150			BUTTERFLY	T06AC61	10	24	BUTTERFLY 600# RF LTCS A352 LCC 316SS TRIM TOV METAL DBL FLG SHORT PAT GO	0	0		
160							*****GATE VALVE*****				
160											
170	U04		GATE	G06AC48	0.75	1.5	GATE 600# RF LTCS A350 LF2 CL.1 TRIM 16 WEDGE HW	0	0		
180	U04		GATE	G06AC41	2	6	GATE 600# RF LTCS A352 LCC TRIM 16 WEDGE HW	0	0		
190	U04		GATE	G06AC42	12	12	GATE 600# RF LTCS A352 LCC TRIM 16 WEDGE GO	0	0		
200			GATE	G06AC42	6	8	GATE 600# RF LTCS A352 LCC TRIM 16 WEDGE GO	0	0		
210							***** GLOBE VALVES *****				
210											
220	U04		GLOBE	L06AC48	0.75	1.5	GLOBE 600# RF LTCS A350 LF2 CL.1 TRIM 16 OS&Y HW	0	0		
230			GLOBE	L06AC41	2	4	GLOBE 600# RF LTCS A352 LCC TRIM 16 OS&Y HW	0	0		
240			GLOBE	L06AC42	6	10	GLOBE 600# RF LTCS A352 LCC TRIM 16 OS&Y GO	0	0		
250							***** CHECK VALVES *****				
250											
260	U04	VC1	CHECK	C06AC50	0.75	1.5	CHECK PISTON 600# RF LTCS A350 LF2 CL.1 TRIM 16	0	0		
270			CHECK	C06AC45	2	8	CHECK DUAL WAFER LUG 600# RF LTCS A352 LCC TRIM 16 THROU BOLTED	0	0		
280			CHECK	C06AC52	10	24	CHECK DUAL DBL FLG 600# RF LTCS A352 LCC TRIM 16	0	0		
290							***** PIPE *****				
290											
300											
300			PIPE	5358512	0.75	1.5	PIPE SCH 160 SMLS LTCS A333 GR.6 PE	0	0		
310			PIPE	5358515	2	2	PIPE SCH 160 SMLS LTCS A333 GR.6 BE	0	0		
320			PIPE	5358514	3	8	PIPE SCH XS SMLS LTCS A333 GR.6 BE	0	0		
330			PIPE	5358507	10	16	PIPE SCH 80 SMLS LTCS A333 GR.6 BE	0	0		
340			PIPE	5358599	18	24	PIPE SCH 80 EFW LTCS A671 CC60 CL.22 BE	0	0		
350		812	PIPE	5358511	0.5	0.5	PIPE SCH XXS SMLS LTCS A333 GR.6 PE	0	0		
360		812	PIPE	5358511	0.75	0.75	PIPE SCH XXS SMLS LTCS A333 GR.6 PE	0	0		
370							***** FITTINGS *****				
370											
380	U04	UK9	ELBOW	5392815	0.5	0.75	ELL BW 90 DEG LR SCH XXS LTCS A420 WPL6	0	0		
390			ELBOW	5392816	0.75	2	ELL BW 90 DEG LR SCH 160 LTCS A420 WPL6	0	0		
400			ELBOW	5392600	3	8	ELL BW 90 DEG LR SCH XS LTCS A420 WPL6	0	0		
410			ELBOW	5392546	10	16	ELL BW 90 DEG LR SCH 80 LTCS A420 WPL6	0	0		
420			ELBOW	5392387	18	24	ELL BW 90 DEG LR SCH 80 LTCS A420 WPL6-W	0	0		
430			ELBOW	5392822	0.75	2	ELL BW 45 DEG LR SCH 160 LTCS A420 WPL6	0	0		
440	U04	UK9	ELBOW	5392831	0.5	0.75	ELL BW 45 DEG LR SCH XXS LTCS A420 WPL6	0	0		
450			ELBOW	5392606	3	8	ELL BW 45 DEG LR SCH XS LTCS A420 WPL6	0	0		
460			ELBOW	5392552	10	16	ELL BW 45 DEG LR SCH 80 LTCS A420 WPL6	0	0		
470			ELBOW	5392391	18	24	ELL 45 DEG SCH 80 LTCS A420 WPL6-W	0	0		
480			TEE	5392825	0.75	2	TEE BW SCH 160 LTCS A420 WPL6	0	0		

	TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES - BUTT WELD PWHT FOR WT >19 MM SEAL WELDING - FOR THREADED CONNECTIONS ONLY PIPE BENDS - NONE THREAD SEALANT - PTFE PASTE	CORR. ALLOW. 3 MM	SOUR SERVICE - NACE MR0175/ ISO 15156 DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.3 FLUID CODES - PIM-SU-5112-TCO	MATERIAL: LTCS FLANGE RATING: 600 RF TEMPERATURE RANGE: -40 / 343 °C	12/16/2024	BY OZAC	U04	Page 1 of 3	600H25

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES
490			TEE	5392609	3	8	TEE BW SCH XS LTCS A420 WPL6	0	0	
500			TEE	5392555	10	16	TEE BW SCH 80 LTCS A420 WPL6	0	0	
510			TEE	5392402	18	24	TEE SCH 80 LTCS A420 WPL6-W	0	0	
520	U04	UK9	TEE	5392879	0.5	0.75	TEE SCH XXS LTCS A420 WPL6	0	0	
530			TEE RED	5392821	0.75	1.5	TEE RED BW SCH 160 LTCS A420 WPL6	1	2	
540			TEE RED	5392469	1.5	2	TEE RED BW SCH XS/160 LTCS A420 WPL6	3	4	
550			TEE RED	5392661	3	6	TEE RED BW SCH XS LTCS A420 WPL6	4	8	
560			TEE RED	5392463	4	8	TEE RED BW SCH 80/XS LTCS A420 WPL6	10	16	
570			TEE RED	5392403	8	8	TEE RED BW SCH 80/XS LTCS A420 WPL6-W	18	20	
580			TEE RED	5392464	10	14	TEE RED BW SCH 80 LTCS A420 WPL6	12	16	
590			TEE RED	5392353	10	20	TEE RED BW SCH 80 LTCS A420 WPL6-W	18	24	
600										
600			CAP	5392826	0.75	2	CAP BW SCH 160 LTCS A420 WPL6	0	0	
610			CAP	5392610	3	8	CAP BW SCH XS LTCS A420 WPL6	0	0	
620			CAP	5392556	10	24	CAP SCH 80 LTCS STL A420 WPL6	0	0	
630										
630			RED CONC	5392827	0.75	1.5	REDUCER CONC SCH 160 LTCS A420 WPL6	1	2	
640			RED CONC	5392462	1.5	2	REDUCER CONC SCH XS/160 LTCS A420 WPL6	3	4	
650			RED CONC	5392458	3	6	REDUCER CONC SCH XS LTCS A420 WPL6	4	8	
660			RED CONC	5392309	4	8	REDUCER CONC SCH 80/XS LTCS A420 WPL6	10	16	
670			RED CONC	5392557	10	14	REDUCER CONC SCH 80 LTCS A420 WPL6	12	16	
680			RED CONC	5392559	10	20	REDUCER CONC SCH 80 LTCS A420 WPL6-W	18	24	
690			RED ECC	5392828	0.75	1.5	REDUCER ECC SCH 160 LTCS A420 WPL6	1	2	
700			RED ECC	5392465	1.5	2	REDUCER ECC SCH XS/160 LTCS A420 WPL6	3	4	
710			RED ECC	5392612	3	6	REDUCER ECC SCH XS LTCS A420 WPL6	4	8	
720			RED ECC	5392311	4	8	REDUCER ECC SCH 80/XS LTCS A420 WPL6	10	16	
730			RED ECC	5392558	10	14	REDUCER ECC SCH 80 LTCS A420 WPL6	12	16	
740			RED ECC	5392313	10	20	REDUCER ECC SCH 80 LTCS A420 WPL6-W	18	24	
750										
750							***** OLETS *****			
760										
760	U04		OLETS	5504460	0.75	1.5	NIPOFLANGE 600# RF SCH 160 LTCS A350 LF2 CL.1	2	2	
770	U04		OLETS	5504461	0.75	2	NIPOFLANGE 600# RF SCH XS/160 LTCS A350 LF2 CL.1	3	8	
780	U04		OLETS	5504462	0.75	2	NIPOFLANGE 600# RF SCH 80/160 LTCS A350 LF2 CL.1	10	24	
790	U04	UJ2	OLETS	5504028	1.5	1.5	NIPOFLANGE 600# RF SCH XS/160 LTCS A350 LF2 CL.1	4	8	
800	U04	UJ2	OLETS	5504029	1.5	1.5	NIPOFLANGE 600# RF SCH 80/160 LTCS A350 LF2 CL.1	10	24	
810	U04		OLETS	5504221	1.5	1.5	NIPOFLANGE 1500# RF SCH XS/160 LTCS A350 LF2 CL.1	3	3	
820										
820	U04		OLETS	5505392	0.75	1	WELDOLET SCH 160 LTCS A350 LF2 CL.1	2	2	
830	U04		OLETS	5505393	0.75	2	WELDOLET SCH XS/160 LTCS A350 LF2 CL.1	3	8	
840	U04		OLETS	5505263	0.75	2	WELDOLET SCH 80/160 LTCS A350 LF2 CL.1	10	24	
850	U04		OLETS	5505335	3	3	WELDOLET SCH XS LTCS A350 LF2 CL.1	8	8	
860	U04		OLETS	5505262	3	8	WELDOLET SCH 80/XS LTCS A350 LF2 CL.1	10	24	
870										
870	U04	UJ2	ELBOFLG	5516659	1.5	1.5	ELBOFLANGE 600# RF SCH XS/160 LTCS A350 LF2 CL.1	4	8	
880	U04	UJ2	ELBOFLG	5516660	1.5	1.5	ELBOFLANGE 600# RF SCH 80/160 LTCS A350 LF2 CL.1	10	24	
890										
890		816					**** FLANGES ****			
900										
900	U04	FA8	TEE RED	5599705	0.5	0.5	FLG SCRDR (FNPT) REDUCING 600# RF A350 LF2 CL.1	0.75	1.5	
910										
910			FLANGE	5851204	0.5	2	FLANGE ADAPTER 600# RF 316 SS X 12 MM TUBE	0	0	
920										
920	U04		FLANGE	5557038	0.5	2	FLG WN 600# RF LTCS A350 LF2 CL.1 SCH 160	0	0	
930	U04		FLANGE	5557036	3	8	FLG WN 600# RF LTCS A350 LF2 CL.1 SCH XS	0	0	
940	U04		FLANGE	5557040	10	24	FLG WN 600# RF LTCS A350 LF2 CL.1 SCH 80	0	0	
950	U04		BLIND	5615598	0.5	24	FLG BLIND 600# RF LTCS A350 LF2 CL.1	0	0	
960	U04		BLIND	5615616	0.75	0.75	FLG BLIND 1500# RF LTCS A350 LF2 CL.1	0	0	
970										
970							***** ORIFICE FLANGES *****			
980										
980	U04		ORIFICE FLG	5624351	2	2	WN ORIF SET 600# RF LTCS A350 LF2 CL.1 SCH 160	0	0	
990	U04		ORIFICE FLG	5624352	3	8	WN ORIF SET 600# RF LTCS A350 LF2 CL.1 SCH XS BORE	0	0	
1000	U04		ORIFICE FLG	5624353	10	24	WN ORIF SET 600# RF LTCS A350 LF2 CL.1 SCH 80 BORE	0	0	
1010										
1010							***** PIPE NIPPLES *****			
1020										
1020		812		0			NIPPLE PBE - CUT FROM PIPE			
1030										
1030							***** GASKETS *****			
1040										
1040			GASKET	5673103	0.5	24	GASKET 600# 316SS SPIRAL WOUND GRAPHITE IIR	0	0	
1050	U04	GA6	GASKET	5673018	3	3	GASKET 600# 316SS KAMMPROFILE GRAPHITE ASME B16.5	0	0	

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1060	U04		GASKET	5672251	0.75	0.75	GASKET 1500# 316SS SPIRAL WOUND GRAPHITE IR	0	0	
1070		B1					***** BOLTING *****			
1080										
1080	U04		BOLTING	5675770	0.5	24	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS	0	0	
1090										
1090		SB6	BOLTING	5675675	10	42	JACK SCREW HEX HEAD A320 GR L7M	0	0	
1100							***** BLINDS AND SPACERS *****			
1100										
1110										
1110			SPEC BLIND	5749587	0.75	8	SPECTACLE BLIND 600# RF LTCS A516-GR70	0	0	
1120										
1120			PADDLE	5750040	10	24	PADDLE BLANK 600# RF LTCS A516-GR70	0	0	
1130			PADDLE	5750251	10	24	PADDLE SPACER 600# RF LTCS A516-GR70	0	0	
1140										

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